

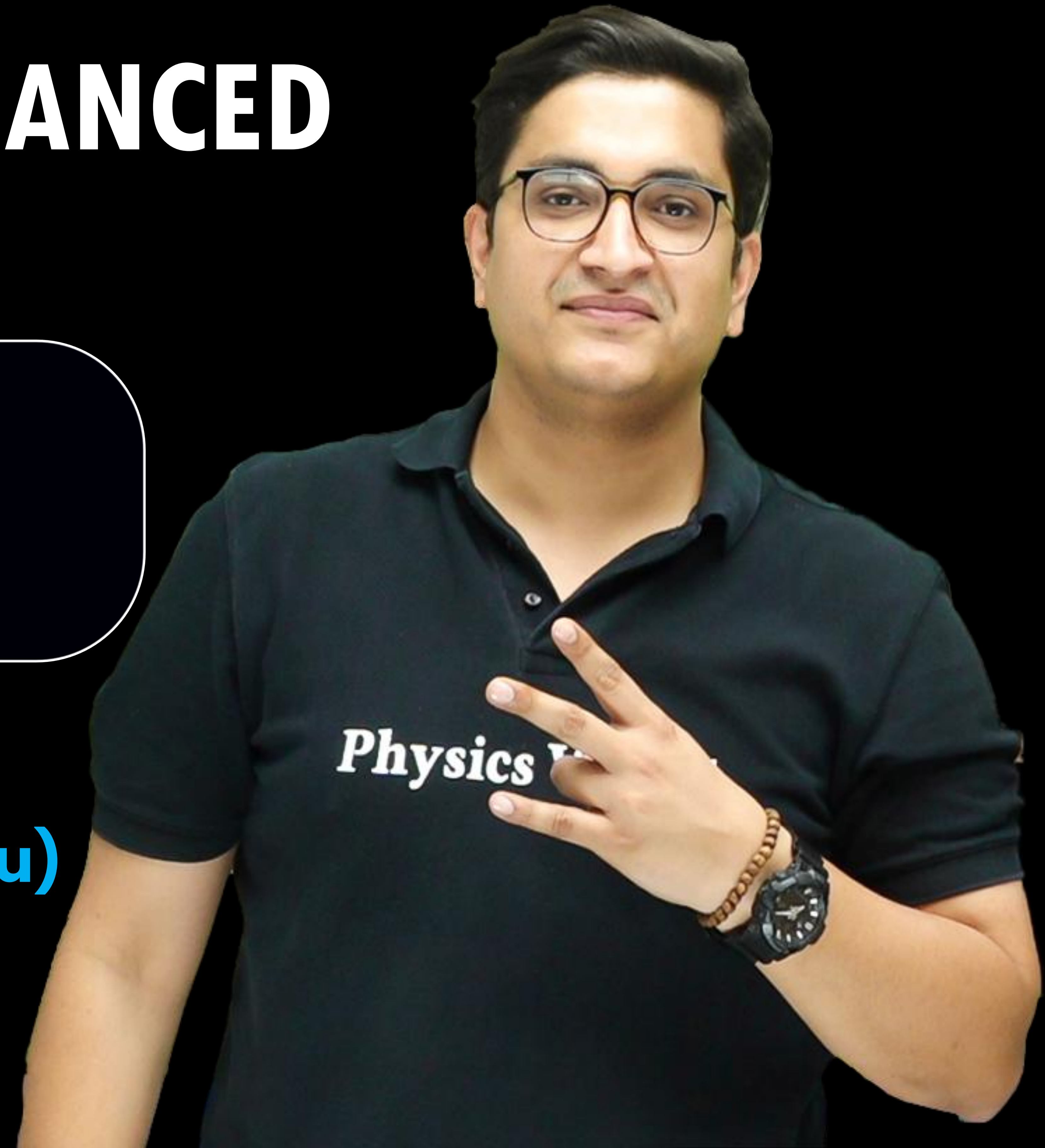
JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

INVERSE TRIGONOMETRIC FUNCTIONS

DIBY (DO IT BY YOURSELF)

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DIBY-01

The value of $\tan^{-1} \left(\frac{\cos\left(\frac{15\pi}{4}\right) - 1}{\sin\left(\frac{\pi}{4}\right)} \right)$ is equal to:

[JEE Mains-2022]

(A) $-\frac{\pi}{4}$

(B) $-\frac{\pi}{8}$

(C) $-\frac{5\pi}{12}$

(D) $-\frac{4\pi}{9}$

[Ans.: (B)]

DIBY-02

Let $x = \sin(2\tan^{-1}\alpha)$ and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$. If $S = \{\alpha \in R: y^2 = 1 - x\}$, then $\sum_{\alpha \in S} 16\alpha^3$ is equal to _____.

[JEE Mains-2022]

[Ans.: 130]

DIBY-03

If the inverse trigonometric functions take principal values,

then $\cos^{-1} \left(\frac{3}{10} \cos \left(\tan^{-1} \left(\frac{4}{3} \right) \right) + \frac{2}{5} \sin \left(\tan^{-1} \left(\frac{4}{3} \right) \right) \right)$ is equal to:

[JEE Mains-2022]

(A) 0

(B) $-\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{6}$

[Ans.: (C)]

DIBY-04

The range of the function $y = \left(\frac{\cos^{-1}(3x-1)}{\pi} + 1 \right)^2$ is

- (A) $[1, 4]$ (B) $[0, \pi]$ (C) $[1, \pi]$ (D) $[0, \pi^2]$

[Ans. : (A)]

DIBY-05

If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, then possible value(s) of a satisfying the equation $x^{100} + y^{100} + z^{100} + \frac{a^2}{x^{50} + y^{50} + z^{50}} = \frac{10a}{3}$ are

- (A) 1 (B) 4 (C) 9 (D) 16

[Ans. : A, C]

DIBY-06

$\operatorname{cosec} \left[2\cot^{-1}(5) + \cos^{-1} \left(\frac{4}{5} \right) \right]$ is equal to :

[JEE Mains-2021]

- (A) $\frac{65}{33}$ (B) $\frac{56}{33}$ (C) $\frac{65}{56}$ (D) $\frac{75}{56}$

[Ans. : (c)]

DIBY-07

$50\tan \left(3\tan^{-1} \left(\frac{1}{2} \right) + 2\cos^{-1} \left(\frac{1}{\sqrt{5}} \right) \right) + 4\sqrt{2}\tan \left(\frac{1}{2}\tan^{-1}(2\sqrt{2}) \right)$ is equal to ____.

[JEE Mains-2022]

[Ans. : 29]

DIBY-08

The value of $\tan \left(2\tan^{-1} \left(\frac{3}{5} \right) + \sin^{-1} \left(\frac{5}{13} \right) \right)$ is equal to:

[JEE Mains-2021]

- (A) $\frac{151}{63}$ (B) $\frac{-291}{76}$ (C) $\frac{220}{21}$ (D) $\frac{-181}{69}$

[Ans. : (c)]

DIBY-09

If for some α, β ; $\alpha \leq \beta$, $\alpha + \beta = 8$ and $\sec^2(\tan^{-1} \alpha) + \operatorname{cosec}^2(\cot^{-1} \beta) = 36$, then $\alpha^2 + \beta$ is _____.

[JEE Mains-2025]

[Ans. : 14]

DIBY-10

If the inverse trigonometric functions take principal values, then

$\tan \left(\left(\sin^{-1} \left(\frac{3}{5} \right) \right) - 2 \cos^{-1} \left(\frac{2}{\sqrt{5}} \right) \right)$ is equal to:

[JEE (Adv.)-2024]

(A) $\frac{7}{24}$

(B) $\frac{-7}{24}$

(C) $\frac{-5}{24}$

(D) $\frac{5}{24}$

[Ans. : (B)]

DIBY-11

Let $\alpha = \tan \left(\frac{5\pi}{16} \sin \left(2\cos^{-1} \left(\frac{1}{\sqrt{5}} \right) \right) \right)$ and $\beta = \cos \left(\sin^{-1} \left(\frac{4}{5} \right) + \sec^{-1} \left(\frac{5}{3} \right) \right)$ where the inverse trigonometric functions take principal values. Then, the equation whose roots are α and β is: [JEE Mains-2022]

(A) $15x^2 - 8x - 7 = 0$

(B) $5x^2 - 12x + 7 = 0$

(C) $25x^2 - 18x - 7 = 0$

(D) $25x^2 - 32x + 7 = 0$

[Ans.: (c)]

DIBY-12

If the domain of the function $f(x) = \cos^{-1} \left(\frac{2x-5}{11-3x} \right) + \sin^{-1} (2x^2 - 3x + 1)$ is the interval $[\alpha, \beta]$, then $\alpha + 2\beta$ is equal to: [JEE Mains-2026]

(A) 5

(B) 1

(C) 3

(D) 2

[Ans.: (c)]

DIBY-13

If the domain of the function $f(x) = \sin^{-1} \left(\frac{5-x}{3+2x} \right) + \frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma) - \{\delta\}$, then $6(\alpha + \beta + \gamma + \delta)$ is equal to

[JEE Mains-2026]

- (A) 68 (B) 70 (C) 66 (D) 67

[Ans.: (B)]

DIBY-14

Let $[.]$ denote that greatest integer function. If the domain of the function

$f(x) = \sin^{-1} \left(\frac{x+[x]}{3} \right)$ is $[\alpha, \beta)$, then $\alpha^2 + \beta^2$ is equal to:

[JEE Mains-2026]

- (A) 2 (B) 5 (C) 10 (D) 13

[Ans.: (B)]



DIBY-15

Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is:

[JEE Mains-2022]

- (A) $\left(-\infty, \frac{1}{4}\right]$ (B) $\left[-\frac{1}{4}, \infty\right)$ (C) $\left(-\frac{1}{3}, \infty\right)$ (D) $\left(-\infty, \frac{1}{3}\right]$

[Ans.: (B)]

DIBY-16

$\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}x\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}x\right)$, $x \neq 0$ is equal to

- (A) x (B) $2x$ (C) $2/x$ (D) None of these

[Ans.: (C)]

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DIBY-17

$\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$ is equal to: (The inverse trigonometric functions take the principal values)

[JEE Mains-2021]

- (A) $3\pi - 11$ (B) $3\pi + 1$ (C) $4\pi - 11$ (D) $4\pi - 9$

[Ans.: (C)]

DIBY-18

If $a = \sin^{-1}(\sin(5))$ and $b = \cos^{-1}(\cos(5))$, then $a^2 + b^2$ is equal to

[JEE Mains-2024]

- (A) $4\pi^2 + 25$ (B) $8\pi^2 - 40\pi + 50$ (C) $4\pi^2 - 20\pi + 50$ (D) 25

[Ans.: (C)]

DIBY-19

Let m and M respectively be the minimum and the maximum values of

$f(x) = \sin^{-1}2x + \sin 2x + \cos^{-1}2x + \cos 2x$, $x \in \left[0, \frac{\pi}{8}\right]$. Then $m + M$ is equal to

[JEE Mains-2024]

- (A) $1 + \sqrt{2} + \pi$ (B) $(1 + \sqrt{2})\pi$ (C) $\pi + \sqrt{2}$ (D) $1 + \pi$

[Ans.: A]

DIBY-20

The number of solutions of the equation $|y| = \cos x$ and $y = \cot^{-1}(\cot x)$ in $\left(-\frac{3\pi}{2}, \frac{5\pi}{2}\right)$

is :

[JEE Mains-2021]

- (A) 2 (B) 4 (C) 6 (D) none of these

[Ans.: (A)]

DIBY-21

If $a\sin^{-1}x - b\cos^{-1}x = c$, then the value of $a\sin^{-1}x + b\cos^{-1}x$ (whenever exists) is equal to :

- (A) 0 (B) $\frac{\pi ab + c(b-a)}{a+b}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi ab + c(a-b)}{a+b}$
[Ans. : (D)]

DIBY-22

The range of $f(x) = \cot^{-1}(-x) - \tan^{-1}x + \sec^{-1}x$ is :

- (A) $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ (B) $\left[\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right]$ (C) $\left(-\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (D) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right)$
[Ans. : (B)]

DIBY-23

Let $f: R \rightarrow \left(0, \frac{\pi}{6}\right]$ be defined as $f(x) = \sin^{-1}\left(\frac{4}{4x^2 - 12x + 17}\right)$ then $f(x)$ is :

- (A) injective as well as surjective (B) surjective but not injective
 (C) injective but not surjective (D) neither injective nor surjective
[Ans. : (B)]

DIBY-24

If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to **[JEE Mains-2019]**

- (A) 7π (B) 10 (C) 0 (D) π

[Ans.: (D)]

DIBY-25

The set of all values of k for which $(\tan^{-1} x)^3 + (\cot^{-1} x)^3 = k\pi^3, x \in R$, is the interval: **[JEE Mains-2022]**

- (A) $\left[\frac{1}{32}, \frac{7}{8}\right)$ (B) $\left(\frac{1}{24}, \frac{13}{16}\right)$ (C) $\left[\frac{1}{48}, \frac{13}{16}\right]$ (D) $\left[\frac{1}{32}, \frac{9}{8}\right)$

[Ans.: A]

DIBY-26

The least integral value of k for which

$(k - 2)x^2 + 8x + k + 4 > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)$ for all $x \in R$, is

- (A) -7 (B) -3 (C) 3 (D) 5

[Ans.: (D)]

DIBY-27

Which of the following is/are true?

(A) $\tan^{-1} \frac{1}{3} = \frac{1}{2} \sin^{-1} \frac{3}{5}$

(B) $\tan^{-1} \frac{1}{3} = \frac{\pi}{4} - \cot^{-1} 2$

(C) $\tan^{-1} \frac{1}{3} = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} \frac{4}{5}$

(D) $\tan^{-1} \frac{1}{3} = \frac{\pi}{4} - \cot^{-1} 3$

[Ans. : A, B, C]

DIBY-28

Using the principal values of the inverse trigonometric functions, the sum of the maximum and the minimum values of $16 \left((\sec^{-1} x)^2 + (\operatorname{cosec}^{-1} x)^2 \right)$ is:

[JEE Mains-2025]

(A) $24\pi^2$

(B) $22\pi^2$

(C) $31\pi^2$

(D) $18\pi^2$

[Ans. : (B)]

DIBY-29

Let the maximum value of $(\sin^{-1} x)^2 + (\cos^{-1} x)^2$ for $x \in \left[-\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}\right]$ be $\frac{m}{n} \pi^2$, where $\gcd(m, n) = 1$. Then $m + n$ is equal to _____.
[JEE Mains-2026]
[Ans. : 65]

DIBY-30

If $\frac{\sin^{-1} x}{a} = \frac{\cos^{-1} x}{b} = \frac{\tan^{-1} y}{c}$; $0 < x < 1$, then the value of $\cos\left(\frac{\pi c}{a+b}\right)$ is: **[JEE Mains-2021]**
 (A) $\frac{1-y^2}{y\sqrt{y}}$ (B) $1 - y^2$ (C) $\frac{1-y^2}{1+y^2}$ (D) $\frac{1-y^2}{2y}$
[Ans. : (C)]

DIBY-31

Considering the principal values of inverse trigonometric functions, the value of the expression $\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right) - 2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$ is equal to: **[JEE Mains-2026]**
 (A) $\frac{16}{63}$ (B) $-\frac{33}{56}$ (C) $-\frac{16}{63}$ (D) $\frac{33}{56}$
[Ans. : (D)]

DIBY-32

Let $g(x) = ax + b$, where $a < 0$ and g is defined from $[1, 3]$ onto $[0, 2]$ then the value of $\cot\left(\cos^{-1}(|\sin x| + |\cos x|) + \sin^{-1}(-|\cos x| - |\sin x|)\right)$ is equal to

- (A) $g(1)$ (B) $g(2)$ (C) $g(3)$ (D) $g(1) + g(3)$

[Ans.: (C)]

DIBY-33

$\cos\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{33}{65}\right)$ is equal to:

[JEE Mains-2025]

- (A) 1 (B) 0 (C) $\frac{32}{65}$ (D) $\frac{33}{65}$

[Ans.: (B)]

DIBY-34

For $n \in \mathbb{N}$, if $\cot^{-1}3 + \cot^{-1}4 + \cot^{-1}5 + \cot^{-1}n = \frac{\pi}{4}$, then n is equal to _____.

[JEE Mains-2024]

[Ans.: 47]

DIBY-35

If $A = \frac{1}{1} \cot^{-1} \left(\frac{1}{1} \right) + \frac{1}{2} \cot^{-1} \left(\frac{1}{2} \right) + \frac{1}{3} \cot^{-1} \left(\frac{1}{3} \right)$ and $B = 1 \cot^{-1}(1) + 2 \cot^{-1}(2) + 3 \cot^{-1}(3)$, then $|B - A|$ is equal to $\frac{a\pi}{b} + \frac{c}{d} \cot^{-1}(3)$, where $a, b, c, d \in \mathbb{N}$ and are in their lowest form, find $(a + b + c + d)$.

[Ans. : 40]

DIBY-36

If $S_n = \sum_{r=1}^n r!$, then for $n > 6$ (given $\sum_{r=1}^6 r! = 873$)

Column-I

Column-II

(A) $\sin^{-1} \left(\sin \left(S_n - 7 \left[\frac{S_n}{7} \right] \right) \right)$

(p) $5 - 2\pi$

(B) $\cos^{-1} \left(\cos \left(S_n - 7 \left[\frac{S_n}{7} \right] \right) \right)$

(q) $2\pi - 5$

(C) $\tan^{-1} \left(\tan \left(S_n - 7 \left[\frac{S_n}{7} \right] \right) \right)$

(r) $6 - 2\pi$

(D) $\cot^{-1} \left(\cot \left(S_n - 7 \left[\frac{S_n}{7} \right] \right) \right)$

(s) $5 - \pi$

(where $[]$ denotes greatest integer function)

(t) $\pi - 4$

[Ans. : (A) $\rightarrow$ (p); (B) $\rightarrow$ (q); (C) $\rightarrow$ (p); (D) $\rightarrow$ (s)]

DIBY-37

If $0 \leq x \leq 1$ and $\theta = \sin^{-1} x + \cos^{-1} x - \tan^{-1} x$, then:

- (A) $\theta \leq \pi/2$ (B) $\theta \geq \pi/4$ (C) $\theta = \pi/4$ (D) $\pi/4 \leq \theta \leq \pi/2$
[Ans.: (D)]

DIBY-38

Range of $f(x) = \cos^{-1} \left(\frac{1+x^2}{2x} \right) + \sqrt{2-x^2}$ is

- (A) $\left\{0, 1 + \frac{\pi}{2}\right\}$ (B) $\{0, 1 + \pi\}$ (C) $\left\{1, 1 + \frac{\pi}{2}\right\}$ (D) $\{1, 1 + \pi\}$
[Ans.: (D)]

DIBY-39

The value of $\sec \left[\sin^{-1} \left(-\sin \frac{50\pi}{9} \right) + \cos^{-1} \cos \left(-\frac{31\pi}{9} \right) \right]$ is equal to:

- (A) $\sec \frac{10\pi}{9}$ (B) $\sec \frac{\pi}{9}$ (C) 1 (D) -1
[Ans.: (D)]

Paragraph:

Let α, β and γ are the roots of the equation $x^3 + 6x + 3 = 0$ and

$$A = \cos^{-1} \left(\sin \left((\alpha + \beta)^{-1} + (\beta + \gamma)^{-1} + (\gamma + \alpha)^{-1} \right) \right)$$

$$B = \cos \left(\tan^{-1} \left(\sin \left(\frac{\alpha + \beta + \gamma}{2} \right) \right) \right); C = \sec^{-1} \left(\operatorname{cosec}((1 - \alpha)(1 - \beta)(1 - \gamma)) \right).$$

1. If the range of the quadratic trinomial $g(x) = x^2 - 2Bx + k$ is $[0, \infty)$, then range of k equals:

(A) $[1, \infty)$ (B) $(1, \infty)$ (C) $\{1\}$ (D) $(-\infty, 1]$
[Ans.: (C)]

2. The value of $(5A + B - C)$ is equal to:

(A) 1 (B) 10 (C) 5 (D) 0
[Ans.: (A)]

3. Range of the function $f(x) = \frac{(5A - C)x^5 + 6Bx^2}{x^4 + (B - 1)x^3 + 1}$, is:

(A) $[3, \infty)$ (B) $[0, 3]$ (C) $[-3, 3]$ (D) $(-\infty, \infty)$
[Ans.: (B)]

DIBY-41

If $\arcsin x + \arcsin y + \arcsin z = \pi$, then prove that: $(x, y, z > 0)$

$$x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2} = 2xyz$$

DIBY-42

If $\sin^{-1} \frac{x}{a} + \sin^{-1} \frac{y}{b} = \sin^{-1} \frac{c^2}{ab}$, show that $b^2x^2 + 2xy\sqrt{a^2b^2 - c^4} + a^2y^2 = c^4$

DIBY-43

If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$, then:

- (A) $x^2 + y^2 + z^2 + 2xyz = 1$
- (B) $2(\sin^{-1} x + \sin^{-1} y + \sin^{-1} z) = \cos^{-1} x + \cos^{-1} y + \cos^{-1} z$
- (C) $xy + yz + zx = x + y + z - 1$
- (D) $\left(x + \frac{1}{x}\right) + \left(y + \frac{1}{y}\right) + \left(z + \frac{1}{z}\right) \geq 6$

[Ans. : A, B]

I.T.F. EQUATIONS



DIBY-44

Solve the equation, $\sin^{-1} \left(\frac{3}{x} \right) + \sin^{-1} \left(\frac{4}{x} \right) = \frac{\pi}{2}$

[Ans. : 5]

DIBY-45

Solve the equation $\tan^{-1} \left(\frac{x-1}{x-2} \right) + \tan^{-1} \left(\frac{x+1}{x+2} \right) = \frac{\pi}{4}$

[Ans. : $\pm \frac{1}{\sqrt{2}}$]

DIBY-46

If $\cos^{-1} x = \tan^{-1} x$, then:

(A) $x^2 = \left(\frac{\sqrt{5}-1}{2} \right)$

(B) $x = \pm \left(\sqrt{\frac{\sqrt{5}-1}{2}} \right)$

(C) $\sin(\cos^{-1} x) = \left(\frac{\sqrt{5}-1}{2} \right)$

(D) $\tan(\cos^{-1} x) = \left(\frac{\sqrt{5}-1}{2} \right)$

[Ans. : A, C]

I.T.F. EQUATIONS



DIBY-47

Find the sum of the squares of all the solution(s) of the equation,

$$2\sin^{-1}(x+2) = \cos^{-1}(x+3)$$

[Ans.: (4)]

DIBY-48

Considering only the principal value of inverse functions, the set

$$A = \left\{x \geq 0 : \tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}\right\}$$

[JEE Mains-2019]

(A) contains two elements

(B) contains more than two elements

(C) is a singleton

(D) is an empty set

[Ans.: (c)]

DIBY-49

The number of solutions of $\tan^{-1}(4x) + \tan^{-1}(6x) = \frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}} < x < \frac{1}{2\sqrt{6}}$, is equal to:

[JEE Mains-2026]

(A) 3

(B) 0

(C) 2

(D) 1

[Ans.: (D)]

I.T.F. EQUATIONS



DIBY-50

The number of real values of x which satisfy $\sin^{-1} \left(\frac{3x}{5} \right) + \sin^{-1} \left(\frac{4x}{5} \right) = \sin^{-1} x$ is equal to: [JEE Mains-2021]

(A) 2

(B) 0

(C) 3

(D) 1

[Ans.: (C)]

DIBY-51

Let $x = \frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos(2\sin^{-1} x) = \frac{1}{9}$ and let α, β ($\alpha > \beta$) be the roots of the equation

$mx^2 - nx - m + n = 0$. Then the point (α, β) lies on the line: [JEE Mains-2024]

(A) $3x + 2y = 2$

(B) $5x - 8y = -9$

(C) $3x - 2y = -2$

(D) $5x + 8y = 9$

[Ans.: (D)]

DIBY-52

Let $x = \sin(2\tan^{-1} \alpha)$ and $y = \sin\left(\frac{1}{2}\tan^{-1} \frac{4}{3}\right)$. If $S = \{\alpha \in \mathbb{R} : y^2 = 1 - x\}$, then $\sum_{\alpha \in S} 16\alpha^3$ is equal to [JEE Mains-2022]

[Ans.: (130)]

I.T.F. EQUATIONS



DIBY-53

For $x \in (-1, 1]$, the number of solutions of the equation $\sin^{-1}x = 2\tan^{-1}x$ is equal to

[JEE Mains-2023]

[Ans.: (2)]

DIBY-54

Let $0 < \alpha < 1$, $\beta = \frac{1}{3\alpha}$ and $\tan^{-1}(1 - \alpha) + \tan^{-1}(1 - \beta) = \frac{\pi}{4}$. Then $6(\alpha + \beta)$ is equal to:

[JEE Mains-2026]

(A) 6

(B) 7

(C) 8

(D) 9

[Ans.: (B)]

DIBY-55

If $\sin(\tan^{-1}(x\sqrt{2})) = \cot(\sin^{-1}\sqrt{1-x^2})$, $x \in (0, 1)$, then the value of x is:

[JEE Mains-2026]

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{2}{3}$

(D) $\frac{5}{8}$

[Ans.: (A)]

I.T.F. EQUATIONS



DIBY-56

If for some α, β ; $\alpha \leq \beta$, $\alpha + \beta = 8$ and $\sec^2(\tan^{-1} \alpha) + \operatorname{cosec}^2(\cot^{-1} \beta) = 36$, then $\alpha^2 + \beta$ is _____.

[JEE Mains-2025]
[Ans.: 14]

DIBY-57

The number of solutions of $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}} < x < \frac{1}{2\sqrt{6}}$, is equal to

(A) 0 (B) 2 (C) 1 (D) 3

[Ans.: (c)]

DIBY-58

For $k \in R$, let the solutions of the equation $\cos(\sin^{-1}(x \cot(\tan^{-1}(\cos(\sin^{-1} x)))) = k$, $0 < |x| < \frac{1}{\sqrt{2}}$ be α and β , where the inverse trigonometric functions take only principal values. If the solutions of the equation $x^2 - bx - 5 = 0$ are $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ and $\frac{\alpha}{\beta}$ then $\frac{b}{k^2}$ is equal to _____

[JEE Mains-2022]
[Ans.: 12]

I.T.F. EQUATIONS



DIBY-59

If $\cos^{-1}\left(\frac{2}{3x}\right) + \cos^{-1}\left(\frac{3}{4x}\right) = \frac{\pi}{2}$, $x > \frac{3}{4}$ then x is equal to

(A) $\frac{\sqrt{145}}{12}$

(B) $\frac{\sqrt{145}}{10}$

(C) $\frac{\sqrt{146}}{12}$

(D) $\frac{\sqrt{145}}{11}$

[JEE Mains-2019]

[Ans.: (A)]

DIBY-60

If $\sin^{-1}\frac{\alpha}{17} + \cos^{-1}\frac{4}{5} - \tan^{-1}\frac{77}{36} = 0$, $0 < \alpha < 13$, then $\sin^{-1}(\sin \alpha) + \cos^{-1}(\cos \alpha)$ is equal to:

(A) π

(B) 16

(C) 0

(D) $16 - 5\pi$

[JEE Mains-2023]

[Ans.: (A)]

DIBY-61

Given the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in $[-1, 1]$ such that $\cos^{-1}x - \sin^{-1}y = \alpha$, $-\frac{\pi}{2} \leq \alpha \leq \pi$.

Then, the minimum value of $x^2 + y^2 + 2xy \sin \alpha$ is

(A) -1

(B) 0

(C) $\frac{-1}{2}$

(D) $\frac{1}{2}$

[JEE Mains-2024]

[Ans.: (B)]

I.T.F. EQUATIONS



DIBY-62

If $k = \tan\left(\frac{\pi}{2} + \frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right) + \tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$, then the number of solutions of the equation $\sin^{-1}(kx - 1) = \sin^{-1}x - \cos^{-1}x$ is _____.

[JEE Mains-2026]

[Ans. : 1]

DIBY-63

Let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, for $x \in R$. Then the number of real solutions of the equation $\sqrt{1 + \cos(2x)} = \sqrt{2}\tan^{-1}(\tan x)$ in the set $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ is equal to:

[JEE (Adv.)-2023]

[Ans. : 3]

DIBY-64

If the total area between the curves $f(x) = \cos^{-1}(\sin x)$ and $g(x) = \sin^{-1}(\cos x)$ on the interval $[-7\pi, 7\pi]$ is A , find the value of $49A$. (Take $\pi = 22/7$)

[Ans. : 3388]

I.T.F. EQUATIONS



DIBY-65

$f(x) = \sin^{-1}(\sin x)$, $g(x) = \cos^{-1}(\cos x)$, then:

(A) $f(x) = g(x)$ if $x \in \left(0, \frac{\pi}{4}\right)$

(B) $f(x) < g(x)$ if $x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

(C) $f(x) < g(x)$ if $\left(\pi, \frac{5\pi}{4}\right)$

(D) $f(x) > g(x)$ if $x \in \left(\pi, \frac{5\pi}{4}\right)$

[Ans. : A, B, C]

DIBY-66

For $0 < A < \frac{\pi}{4}$, value of $\tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \tan^{-1}(\cot A) + \tan^{-1}(\cot^3 A)$ is _____.

[Ans. : π]

DIBY-67

How many ordered solutions (x, y) in positive integers does the equation

$\tan^{-1} x + \cos^{-1} \frac{y}{\sqrt{1+y^2}} = \sin^{-1} \frac{3}{\sqrt{10}}$ have?

[Ans. : 2]

I.T.F. EQUATIONS



DIBY-68

If x and y are positive integer satisfying $\tan^{-1} \left(\frac{1}{x} \right) + \tan^{-1} \left(\frac{1}{y} \right) = \tan^{-1} \left(\frac{1}{7} \right)$, then find the number of ordered pairs of (x, y) .

[Ans. : 6]

DIBY-69 (Feel of Substitution)

Considering the principal values of the inverse trigonometric functions,

$\sin^{-1} \left(\frac{\sqrt{3}}{2} x + \frac{1}{2} \sqrt{1 - x^2} \right)$, $-\frac{1}{2} < x < \frac{1}{\sqrt{2}}$, is equal to

- (A) $\frac{\pi}{4} + \sin^{-1} x$ (B) $\frac{\pi}{6} + \sin^{-1} x$ (C) $\frac{-5\pi}{6} - \sin^{-1} x$ (D) $\frac{5\pi}{6} - \sin^{-1} x$

[Ans. : B]

DIBY-70

The value of x satisfying the equation

$(\sin^{-1} x)^3 - (\cos^{-1} x)^3 + (\sin^{-1} x)(\cos^{-1} x)(\sin^{-1} x - \cos^{-1} x) = \frac{\pi^3}{16}$ can not be equal

to:

- (A) $\cos \frac{\pi}{5}$ (B) $\cos \frac{\pi}{4}$ (C) $\cos \frac{\pi}{8}$ (D) $\cos \frac{\pi}{12}$

[Ans. : A, B, D]

I.T.F. Series



DIBY-71

If S is the sum of the first 10 terms of the series $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + \tan^{-1}\left(\frac{1}{21}\right) + \dots$, then $\tan(S)$ is equal to **[JEE Mains-2020]**

(A) $-\frac{6}{5}$

(B) $\frac{5}{11}$

(C) $\frac{10}{11}$

(D) $\frac{5}{6}$

[Ans. : (D)]

DIBY-72

The value of $\cot\left(\sum_{n=1}^{50} \tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is **[JEE Mains-2022]**

(A) $\frac{26}{25}$

(B) $\frac{25}{26}$

(C) 52

(D) 52

[Ans. : (A)]

DIBY-73

The value of $\lim_{n \rightarrow \infty} 6 \tan\left\{\sum_{r=1}^n \tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right\}$ is equal to **[JEE Mains-2022]**

(A) 1

(B) 2

(C) 3

(D) 6

[Ans. : (C)]

DIBY-74

$\lim_{n \rightarrow \infty} \tan \left\{ \sum_{r=1}^n \tan^{-1} \left(\frac{1}{1+r+r^2} \right) \right\}$ is equal to

[JEE Mains-2021]

[Ans. : 1]

DIBY-75

If $\frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1} \left(\frac{2^{p-1}}{1+2^{2p-1}} \right) = \alpha$, then $\tan \alpha$ is equal to

[JEE Mains-2026]

[Ans. : 2048]

DIBY-76

If $\tan^{-1} \frac{1}{1+2} + \tan^{-1} \frac{1}{1+2.3} + \tan^{-1} \frac{1}{1+3.4} + \dots + \tan^{-1} \frac{1}{1+n(n+1)} = \tan^{-1} \theta$, then θ is equal to:

(A) $\frac{n}{n+2}$

(B) $\frac{n}{n+1}$

(C) $\frac{n+1}{n}$

(D) n

[Ans. : (A)]

DIBY-77

If $\sum_{n=1}^{\infty} \tan^{-1} \frac{8n}{n^4 - 2n^2 + 5} = \pi - \cot^{-1} k$. Find k .

[Ans. : 2]

DIBY-78

The sum of series $\cot^{-1} \left(\frac{9}{2} \right) + \cot^{-1} \left(\frac{33}{4} \right) + \cot^{-1} \left(\frac{129}{8} \right) + \dots \infty$

- (A) $\cot^{-1} \left(\frac{1}{2} \right)$ (B) $\tan^{-1} (2)$ (C) $\cot^{-1} (2)$ (D) $\cot^{-1} (3)$

[Ans. : (C)]

DIBY-79

The sum of the infinite series $\cot^{-1} \left(\frac{7}{4} \right) + \cot^{-1} \left(\frac{19}{4} \right) + \cot^{-1} \left(\frac{39}{4} \right) + \cot^{-1} \left(\frac{67}{4} \right) + \dots$ is:

[JEE Mains-2025]

- (A) $\frac{\pi}{2} + \tan^{-1} \left(\frac{1}{2} \right)$ (B) $\frac{\pi}{2} - \cot^{-1} \left(\frac{1}{2} \right)$ (C) $\frac{\pi}{2} + \cot^{-1} \left(\frac{1}{2} \right)$ (D) $\frac{\pi}{2} - \tan^{-1} \left(\frac{1}{2} \right)$

[Ans. : (D)]

DIBY-80

Find the number of solutions of the equation $\tan \left(\sum_{r=1}^5 \cot^{-1}(2r^2) \right) = \frac{5x+6}{6x+5}$

[Ans. : 0]

DIBY-81

If $\sum_{r=1}^{10} \tan^{-1} \left(\frac{3}{9r^2+3r-1} \right) = \cot^{-1} \left(\frac{m}{n} \right)$ (where m and n are coprime)

then find $(2m + n)$.

[Ans. : 32]

DIBY-82

If $\alpha = \sum_{n=1}^{\infty} \tan^{-1} \left\{ \frac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16} \right\}$ and $\sin \alpha = \frac{a}{b}$ where $a, b \in N$, then find the least value of $a^2 + b^2$.

[Ans. : 41]

DIBY-83

The value of $\sec^{-1} \left(\frac{1}{4} \sum_{k=0}^{10} \sec \left(\frac{7\pi}{12} + \frac{k\pi}{2} \right) \sec \left(\frac{7\pi}{12} + \frac{(k+1)\pi}{2} \right) \right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4} \right]$ equals.

[JEE (Adv.)-2019]
[Ans. : 0]

DIBY-84

Let $x = \cot^{-1} \left(\frac{31}{25} \right) + \cot^{-1} \left(\frac{91}{25} \right) + \cot^{-1} \left(\frac{201}{25} \right) + \cot^{-1} \left(\frac{361}{25} \right) + \dots \infty$, then $15 \cot x$ is _____.

[Ans. : 3]

DIBY-85

Let $S_n = \sum_{n=1}^n \sin^{-1} \left[\frac{(2n+1)}{n(n+1)(\sqrt{n(n+2)} + \sqrt{(n+1)(n-1)})} \right]$.

Find the value of $100 \cos (S_{99})$.

[Ans. : 1]

DIBY-86

Let $S_n = \cot^{-1} \left(3x + \frac{2}{x} \right) + \cot^{-1} \left(6x + \frac{2}{x} \right) + \cot^{-1} \left(10x + \frac{2}{x} \right) + \cdots + n$ terms, where $x > 0$.

If $\lim_{n \rightarrow \infty} S_n = 1$, then x equals:

(A) $\pi/4$

(B) 1

(C) $\tan 1$ (D) $\cot 1$

[Ans. : D]

FEEL OF SUBSTITUTION

DIBY-87

If $f(x) = \tan^{-1} \left(\frac{\sqrt{3}x-3x}{3\sqrt{3}+x^2} \right) + \tan^{-1} \left(\frac{x}{\sqrt{3}} \right)$, $0 \leq x \leq 3$, then :

- (A) the least value $f(x)$ is $-\frac{\pi}{3}$ (B) the greatest value of $f(x)$ is $\frac{\pi}{4}$
 (C) the least value of $f(x)$ is 0 (D) the greatest value of $f(x)$ is $\frac{\pi}{3}$

[Ans. : B, C]

DIBY-88

The number of solution of the equation $2\sin^{-1} \left(\frac{2x}{1+x^2} \right) - \pi x^3 = 0$ is equal to:

- (A) 0 (B) 1 (C) 2 (D) 3

[Ans. : (D)]

DIBY-89

If $f(x) = \sin^{-1} \left[\frac{x^2(2x^2+1)-1}{1+x^2} \right] = \begin{cases} a\pi + b\cos^{-1} x, & 0 \leq x \leq 1 \\ p\pi + q\sin^{-1} x, & -1 \leq x < 0 \end{cases}$, then find the absolute of $(a + b + p + q)$.

[Ans. : 4]

FEEL OF SUBSTITUTION

DIBY-90 (Paragraph for Question Nos. 90 (A) and 90 (B))

Consider, $f(x) = \tan^{-1} \left(\frac{\sqrt{1-4x^2}-2\sqrt{3}x}{\sqrt{3-12x^2+2x}} \right)$

(A) If $x \in \left(-\frac{\sqrt{3}}{4}, \frac{1}{2} \right)$ then range of $f(x)$ is :

(A) $\left[0, \frac{\pi}{6} \right]$

(B) $\left[-\frac{\pi}{3}, \frac{\pi}{6} \right)$

(C) $\left[-\frac{\pi}{3}, 0 \right)$

(D) $\left[-\frac{\pi}{3}, \frac{\pi}{2} \right)$

[Ans. : (D)]

(B) The value of $f\left(\frac{\sqrt{3}+1}{4\sqrt{2}}\right) + f\left(\frac{1}{2\sqrt{2}}\right)$ is equal to:

(A) $-\frac{5\pi}{12}$

(B) $-\frac{\pi}{12}$

(C) $-\frac{\pi}{3}$

(D) $\frac{5\pi}{12}$

[Ans. : (C)]

NUMBER OF SOLUTION (GRAPHICALLY)

DIBY-91

If $f(x) = \tan^{-1} \left(\sqrt{\frac{1-\cos 2}{1+\cos 2}} + x^2 \right)$ and $g(x) = \begin{cases} |x| & x \neq 0 \\ 1 & x = 0 \end{cases}$ then number of solution of the equation $f(x) = g(x)$ is (are) :

- (A) 0 (B) 1 (C) 2 (D) 3

[Ans. : (D)]

DIBY-92

The number of non zero roots of the equation $\sqrt{\sin x} = \cos^{-1}(\cos x)$ in $(0, 2\pi)$:

- (A) 0 (B) 1 (C) 2 (D) Infinite

[Ans. : (B)]

*It is never about end result
It's always about Journey*



THANK
You

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